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Orchestrating Hybrid Quantum-Classical Workflows with IBM LSF

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Why does this matter?

Plus ça change … Schedulers have always dealt with diverse environments

1990s

CPUs

Mixed-vendor UNIX clusters — schedulers managing multiple architectures

2010s

GPUs

Accelerators arrive — scheduling now spans CPU + co-processor

2020s

QPUs

Quantum arrives as a complementary resource

From heterogeneous clusters to GPUs and … now QPUs

How did we get here?

QRMI, jointly developed by IBM and STFC, provides an open-source, platform- and scheduler-independent interface for managing quantum resources. More information on QRMI here https://quantum.cloud.ibm.com/docs/en/guides/qrmi

Demonstrated across major HPC schedulers—available open-sourced LSF plugins, showing true scheduler-agnostic interoperability

LSF plugins available here https://github.com/IBM/lsf-quantum

Hybrid quantum–classical workflows require tight coordination across CPUs, GPUs, and QPUs, making HPC schedulers essential for end-to-end orchestration

Rising heterogeneity in HPC + QPUs introduces new middleware and synchronization challenges that demand a neutral, extensible resource layer

Strong global momentum (e.g., ORNL QCUP, Q-DESSI) is pushing toward quantum-centric supercomputing and unified hybrid infrastructure

The workload & mapping it to LSF

Sample-based Quantum Diagonalization (SQD)

- Estimate the ground-state energy of a molecule — the lowest-energy state of its electrons
- Shipped as part of IBM Qiskit add-ons
- Run at scale at RIKEN on Fugaku and IBM QPUs
- The quantum circuit samples electronic configurations; classical compute used for intensive linear algebra and post-processing

Mapping SQD to LSF

- Monolithic SQD script → four LSF jobs with *done* dependencies

MAP
Inputs -> quantum problem

↓

TRANSPILE
Optimize circuit for target QPU

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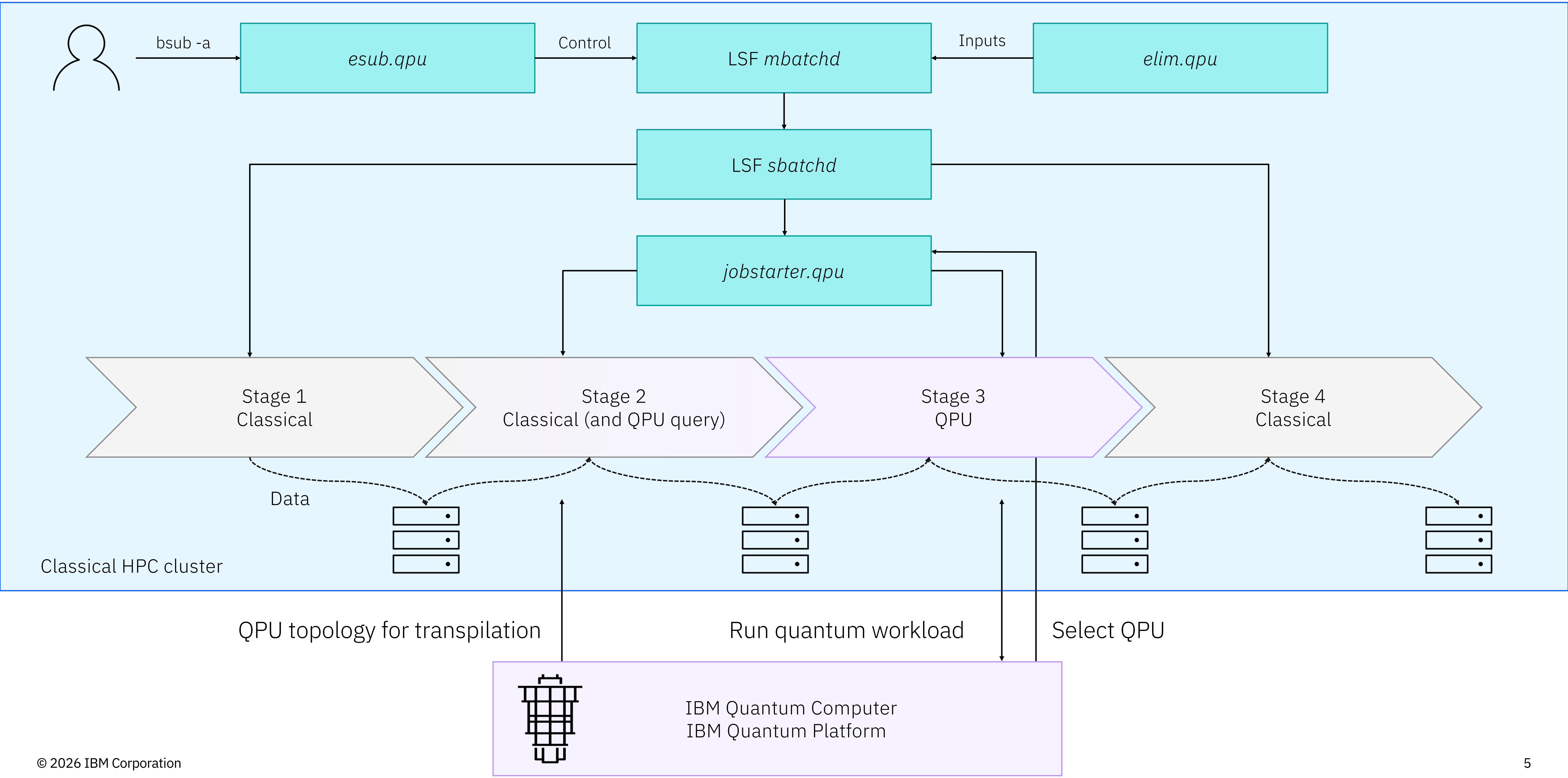
RUN ON QPU
Execute via Qiskit primitives

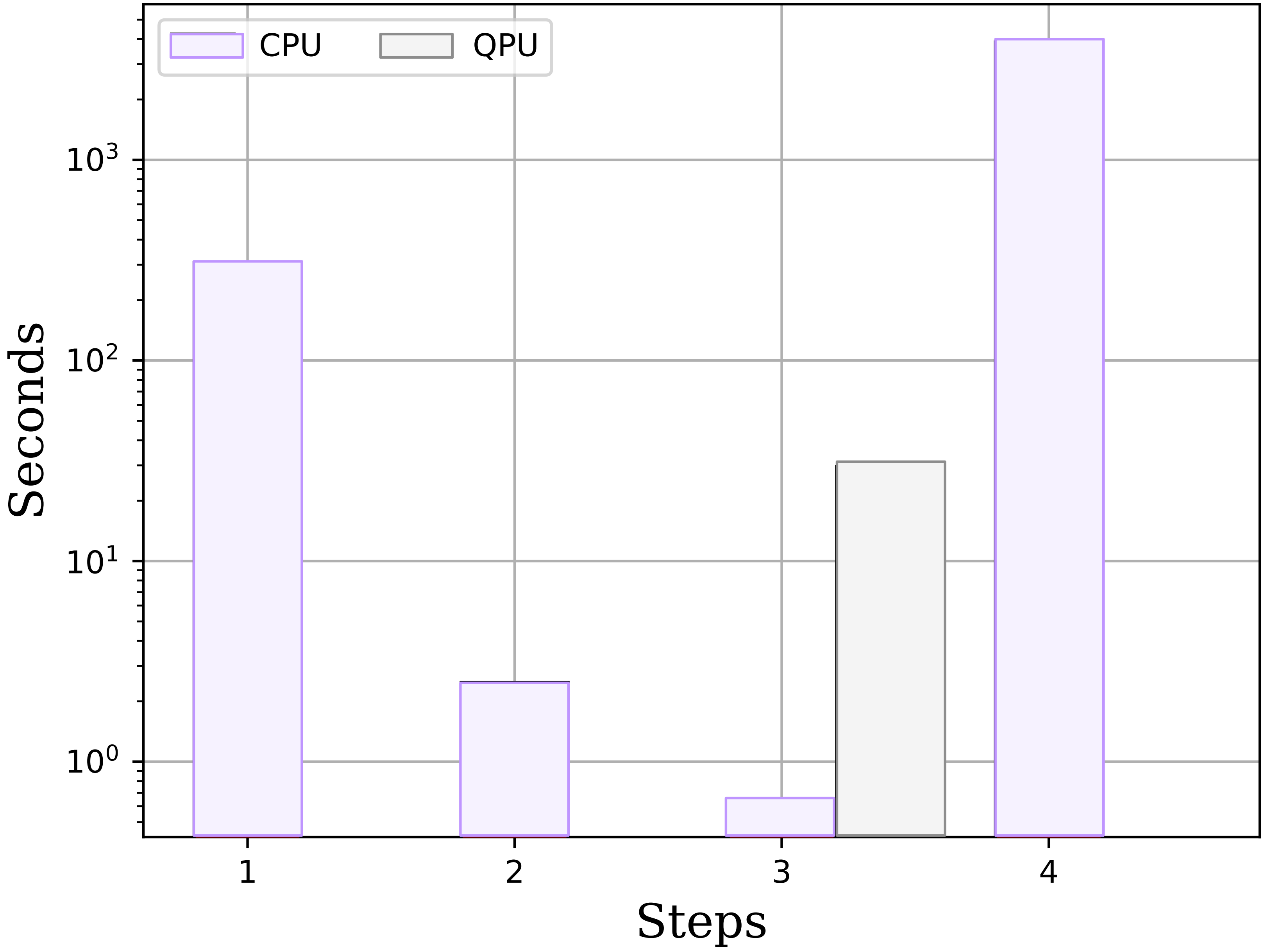
↓

POST PROCESS
Return classical result (energy)

Running Hybrid Quantum-Classical Workflows with IBM LSF



SQD CPU and QPU times



Discussion

Accounting — What resource (QPU) was consumed, was it used well, how much did it cost?

Task pending times — estimate of queue depth and pending time for QPU tasks

Topology — QPU topology, coherency, error rates etc.

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